Model Building Pipeline

Two types of ML Algorithms :-

1. Supervised ML Algo.
2. Unsupervised ML Algo.

- Supervised Algo. :- Works with labelled data.
  - I/P (features)
  - O/P (target / label)

- There can be multiple features
- These features helps in calculating / predicting the target.
- Ex- House Price Prediction
  - ↳ No. of Rooms
  - ↳ Area in sq. ft.
  - ↳ House Age
  - ⋮

  } features ↓ Depending on these features, we can predict the house price

∴ House Price → Target / Output

To solve such kind of problems (above one), we use ML Algorithms

These ML algorithms works with two types of problems / tasks.

- Regression Problem
- Classification Problem

- Classification Problem

★ Regression :- It is used when the target is numerical.
↓
(continuous value)

Ex- House Price, Car Price, Salary, etc.

★ Classification :- It is used when the target is categorical
↓
(a label / a class)

Ex- Email detection, ⇝ [spam / not spam]
Cancer prediction, ⇝ [cancer / no cancer]
Exam Evaluation, etc. ⇝ [pass / fail]

Classification problems are divided into two types :-

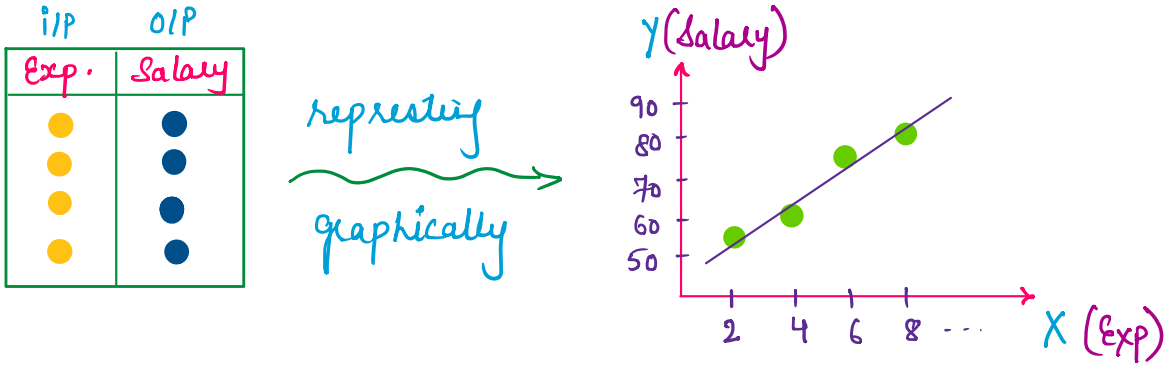
- Binary Classification → Having two classes / labels
- Multiclass " → " more than two classes

Algorithms used with both the problems :-

Regression	Classification
Linear Regression	Logistic Regression
KNN	KNN
Decision Tree	Decision Tree
Bagging	Bagging
Boosting	Boosting
	Naive Bayes
	SVM

Regression :-

Ex- Salary Prediction



- We use a mathematical eq. / logic to predict salary based on experience.
- As we have multiple features, we cannot represent the problem graphically but we can solve it theoretically & practically

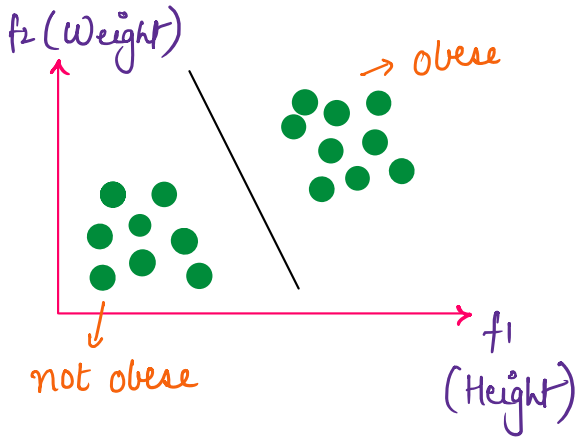
* Classification :-

Ex- Email Detection ⇝ spam / not-spam

[two classes]

(f1) Height	(f2) Weight	Label
●	●	obese
●	●	not obese
●	●	⋮
●	●	
●	●	
●	●	

representing graphically



* Model Building Pipeline :-

Pipeline :- Indicates flow of something / task executed step by step

Model Building :-

↳ Creating ML model

↳ means to train the model on huge amount of data

↳ using an algo. & feeding data inside it

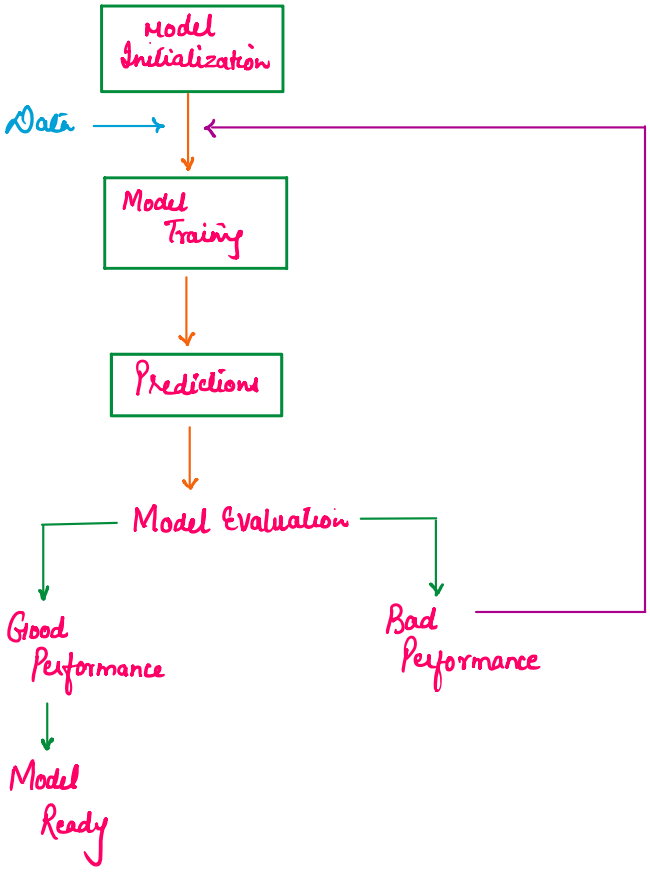
↓

using / applying maths

So,

Model Building Pipeline :-

- Create a model
- Feed the data
- Take predictions
- Evaluation



Before feeding the data into the model, we have to check the data

↓

to see whether the data is perfect for ML algo.

To make the data perfect → Do Data Preprocessing

★ Data Pre-processing :-

Data ⇝ raw data

↓

preprocess the raw data

Data pre-processing steps :-

- Missing Value Detection & Treatment
- Outlier Detection & Treatment

- Relationship b/w Multiple Variables
- Encoding the categorical columns
- Feature Engineering
- Scaling the data

Qual.	Exp.	Salary
MBA	●	●
BTech	●	●
BCA	●	●
⋮	⋮	⋮

categorical values

∴

Data pre-processing means transforming the raw data into a clean & structured format for prediction or analysis.

* Data Splitting :-

After model training → evaluate model's performance

↓

Same Dataset

↳ Rely on this

for a good model → We test the model's performance on some other set of data

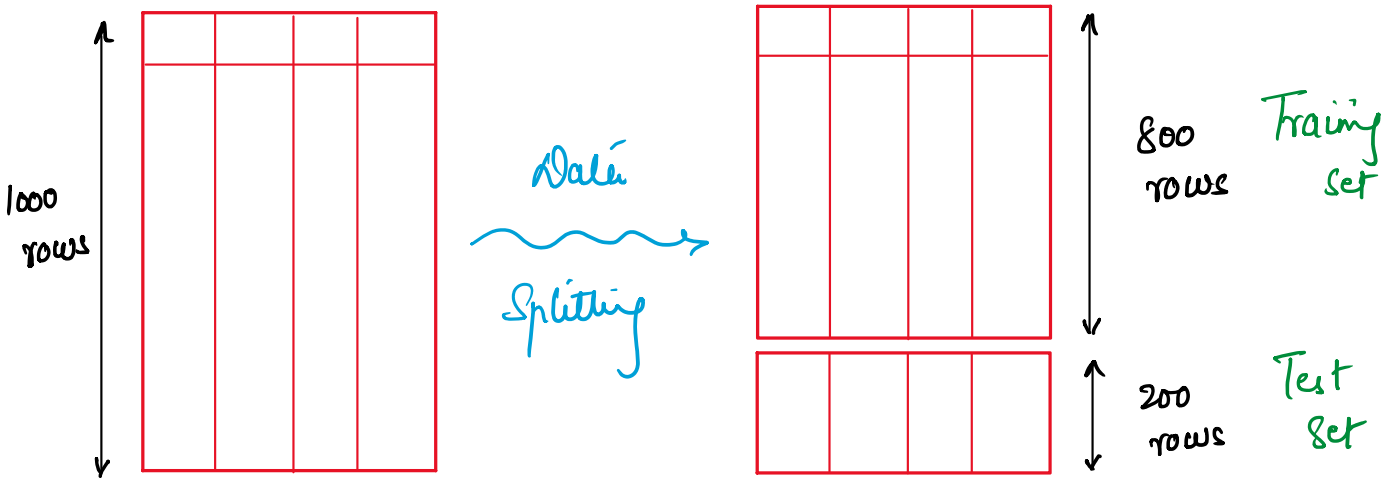
Ques. from where to get this other data which is similar?

Solution → Data Splitting

Dataset → divide it into two parts

① Training data

② Test data


1000 rows
Data
Splitting
800 rows
Training set
200 rows
Test set